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Principal component analysis of molecular dynamics: On the use of Cartesian vs. internal coordinates

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Principal component analysis of molecular dynamics simulations is a popular method to account for the essential dynamics of the system on a low-dimensional free energy landscape. Using Cartesian coordinates, first the translation and overall rotation need to be removed from the trajectory. Since the rotation depends via the moment of inertia on the molecule's structure, this separation is only straightforward for relatively rigid systems. Adopting millisecond molecular dynamics simulations of the folding of villin headpiece and the functional dynamics of BPTI provided by D. E. Shaw Research, it is demonstrated via a comparison of local and global rotational fitting that the structural dynamics of flexible molecules necessarily results in a mixing of overall and internal motion. Even for the small-amplitude functional motion of BPTI, the conformational distribution obtained from a Cartesian principal component analysis therefore reflects to some extend the dominant *overall* motion rather than the much smaller *internal* motion of the protein. Internal coordinates such as backbone dihedral angles, on the other hand, are found to yield correct and well-resolved energy landscapes for both examples. The virtues and shortcomings of the choice of various fitting schemes and coordinate sets as well as the generality of these results are discussed in some detail. © *2014 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4885338]

I. INTRODUCTION

Classical molecular dynamics (MD) simulations allow us to study the structure and dynamics of molecular systems in microscopic detail. To obtain a concise but correct interpretation of the ever-growing amount of simulation data, numerous analysis strategies have been developed, including network theory and Markov state models,[1–5] stochastic approaches such as the Langevin equation,[6–9] and a variety of dimensionality reduction methods.[10–17] The latter attempt to reduce the description of the highly correlated molecular motion of $3N$ atomic coordinates to a few collective degrees of freedom x, that account for the essential dynamics of the system. A simple and widely used technique is the principal component analysis (PCA), which represents a linear transformation that diagonalizes the covariance matrix and thus removes the instantaneous linear correlations among the coordinates.[18] Ordering the eigenvalues of the transformation decreasingly, it has been shown that a large part of the system's fluctuations can be described by the first few principal components.[11,19–24] The resulting low-dimensional representation of the dynamics can then be used to construct the free energy landscape $\Delta G(x)$, which reveals the metastable conformational states (the minima of ΔG) and the transition states (the barriers of ΔG) of the system.[25–27]

Commonly a PCA of a MD simulation is performed on the mass-weighted Cartesian coordinates of the molecule. To obtain the internal motion of the system, first its translation and overall rotation need to be removed from the trajectory. This procedure, usually referred to as "rotational fit," is well established in the case that the molecule is almost rigid, but less so for flexible systems.[28–31] To circumvent this problem, internal coordinates such as contacts and inter-atom distances[32–34] or the backbone dihedral angles[23,35–41] (e.g., ϕ_n, ψ_n) may be used. In fact, by calculating the free energy landscape from MD simulations of various small peptides and RNA, it has been found that results from a PCA using backbone dihedral angles (henceforth termed "dPCA") may provide a qualitatively different picture of the energy landscape than corresponding results from a PCA using Cartesian coordinates (henceforth termed "cPCA").[23,38,39] Since the results of the dPCA were reconfirmed by independent dynamical clustering studies, this apparent failure of the cPCA was attributed to the incomplete separation of overall and internal motion. Surprisingly, however, a similar finding was also obtained for a quite rigid RNA hairpin which should not pose a separation problem,[39] showing that the underlying physics is not well understood. Moreover, large variances in dihedral space (as detected by a dPCA) do not necessarily correspond to large conformational changes in the Cartesian space of interest. Hence it seems questionable if the advantageous features of the dPCA found for small peptides also apply to the study of protein dynamics.

In this work, we wish to clarify the above explained issues on the use of Cartesian and internal coordinates for a PCA of protein dynamics. To this end, we adopt long (≈ 1 ms) MD simulations of the folding of villin headpiece protein[42] (HP35) and the native state of bovine pancreatic trypsin inhibitor[43] (BPTI) previously performed by D. E. Shaw research. The former system is a standard example of protein

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folding showing large-amplitude motion and therefore should represent a separation problem for an analysis based on Cartesian coordinates. The latter is a relatively rigid system that exhibits small-amplitude functional motion, for which no such problems are expected.

To explain the origin and impact of the separation problem for a PCA, we start with the structural analysis of a small part of the system (say, a single residue) and also perform the rotational fit only for this part. Since for a small system the inertia tensor changes only little upon a (ϕ, ψ) conformational transition, both dPCA and cPCA give the same results.[37] In a second step, we increase the size of the considered subsystem for the PCA as well as for the rotational fit, in order to learn at what point and why the problem arises. The analysis shows that the above mentioned artifacts of the cPCA energy landscape are a consequence of the inability of the rotational fitting procedure to correctly and completely remove the overall rotation. Interestingly, the fitting problem affects to some extent also the cPCA of the small-amplitude functional dynamics of BPTI, which also explains previous findings.[39] On the other hand, the dPCA is found to yield correct and well-resolved energy landscapes for both studied systems. The comparison of various fitting schemes and coordinate sets provides a clear picture of their applicability, virtues, and shortcomings in the analysis of structural dynamics.

II. THEORY AND METHODS

A. Separation of overall and internal motion

Considering a molecule containing N atoms with mass m_i and Cartesian coordinates $\{\mathbf{r}_i\}$, the translation can be separated exactly by requiring that the molecule's center of mass is fixed, e.g., at the origin of the coordinate system, $\sum_i^N m_i \mathbf{r}_i = 0$. To separate the overall rotation, we decompose the atomic velocities as $\mathbf{v}_i = \mathbf{u}_i + \boldsymbol{\omega} \times \mathbf{r}_i$, where $\mathbf{u}_i$ and $\boldsymbol{\omega}$ are the vibrational and angular velocity, respectively.[29] Hence the kinetic energy of the molecule reads

$$T = \frac{1}{2}\sum_{i=1}^{N} m_i \left[\mathbf{u}_i^2 + (\boldsymbol{\omega} \times \mathbf{r}_i)^2 + \boldsymbol{\omega} \cdot (\mathbf{r}_i \times \mathbf{u}_i)\right] = T_V + T_R + T_{RV}, \quad (1)$$

comprising the vibrational energy T_V, the rotational energy T_R, and the interaction between vibration and overall rotation, the Coriolis energy T_{RV}. To remove the overall rotation, one may perform a rotation of all coordinates of the molecule

$$\mathbf{r}_i' = \mathbf{R}\mathbf{r}_i, \quad (2)$$

where $\mathbf{R}$ is a 3×3 matrix. It is clear, however, that the choice of $\mathbf{R}$ is not unique for a non-rigid molecule and that, in general, the transformation does not completely decouple internal and rotational motion. (One cannot remove a nonlinear coupling T_{RV} by a linear transformation $\mathbf{R}$.) The solution to this problem is well established in the case that the molecule is almost rigid, that is, if the vibrational dynamics is well described by a small-amplitude motion around a single equilibrium structure $\bar{\mathbf{r}}$. In a classic paper, Eckart showed that the rotation $\mathbf{R}$ corresponds to a transformation to the molecular-fixed coordinate system (the "Eckart frame"), which minimizes the Coriolis coupling.[28,31] This procedure is equivalent[44] to a rotational least-square fit proposed by McLachlan,[30] which is commonly used to superpose molecular structures. Here, the overall rotation matrix in Eq. (2) is determined by minimizing the least-square distance between the mass-weighted instantaneous structures $\mathbf{r}' = \{\mathbf{R}^T\mathbf{r}_i\}$ and reference structures $\bar{\mathbf{r}}$.

In the case of large-amplitude motion (such as protein folding), the separation of the overall rotation of the system is less straightforward, since rotational and internal motion are intrinsically coupled via T_{RV}. This is also obvious from the consideration that the total angular momentum $\mathbf{L}$ depends via $\mathbf{L} = \mathbf{I} \cdot \boldsymbol{\omega}$ on the inertia tensor $\mathbf{I}$ of the molecule, which is a function of the molecular coordinates and changes with the internal motion. The problem can be circumvented for an isolated system (e.g., in the gas phase or when the solvent is treated implicitly), because in this case the total angular momentum is conserved and can be set to zero through the initial condition of the trajectory. However, even for overall zero-angular momentum a flexible system can rotate when the angular momentum associated with the rotation is canceled by the angular momentum due to the motions of the individual particles[45] (as is also well established for the rotations of divers, gymnasts and cats[46]).

Another remedy is to change to some internal coordinates $\mathbf{q} = \{q_k\}$ $(k = 1, \ldots, 3N-6)$ of the system. For example, for a simple linear chain of N atoms, we may introduce $N-1$ bond stretch vibrations, $N-2$ bend vibrations, and $N-3$ dihedral angles. (We note in passing, however, that the motion observed by internal coordinates is generally also affected by the overall rotation of the system. For example, consider two rotating particles connected by a harmonic spring. The centrifugal force of the rotation will cause a shift of the equilibrium distance of the harmonic oscillator, thus giving rise to an effective potential.) Nonetheless, in many cases of practical interest it would be more convenient to perform the analysis in Cartesian coordinates. With this end in mind, a number of methods have been suggested to find the "best" rotational fit.[47–56] However, the quality of the resulting description of structure and dynamics may be difficult to assess in practice.

To summarize, the separation of overall and internal motion of a flexible molecule is in general a nontrivial problem, because according to Eq. (1) rotational and internal motion are intrinsically coupled. The level of difficulty to achieve the separation depends on several issues:

(i) The flexibility of the system and its internal and rotational energy. These conditions result in motion with either a smaller or a larger amplitude, the former being a well-established problem, the latter in general is an unsolved one.
(ii) Interaction with the environment. In the case of an isolated system, the total angular momentum is conserved and can be set to zero, which in a simple way eliminates most (but not all[45]) rotation of the system.
(iii) The coordinates used. Internal coordinates are for the most part independent of the rotation, while the use of Cartesian coordinates necessitates some kind of

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rotational fit [cf. Eq. (2)], which requires defining a reference structure and a rotation matrix.

(iv) The desired observable. Highly averaged quantities such as the total internal energy[57] or orientationally averaged NMR relaxation data[58,59] are easier to obtain than detailed structural and energetic information as contained in the multidimensional free energy landscape.

B. Molecular dynamics simulations

Villin Headpiece: HP35 is a 35-residue mini protein with a hydrophobic core, consisting of three helices (residues 4-10, 15-19, and 23-32) that are connected via two turns. To study the folding of HP35, extensive all-atom equilibrium MD simulations of wild-type HP35 and its mutants were carried out by Piana *et al.*[42] at various temperatures, employing the Amber ff99SB*-ILDN force field[60–62] and the TIP3P explicit water.[63] For the present analysis, a $\approx$300 μs long trajectory of the fast folding double mutant (HP35 NleNle) at 380 K was adopted. The trajectory comprises 140 folding and unfolding events within a total of $\approx 1.5 \times 10^6$ snapshots taken at every 200 ps.

Bovine Pancreatic Trypsin Inhibitor: BPTI is a well-studied 58-residue protein that contains a 3_{10} helix (residues 3-6), two $\beta-$sheets (residues 18-24 and 29-35), and an α-helix (residues 48-55), which are connected via three turns and stabilized by three disulfide bonds. To study the functional dynamics of BPTI, Shaw *et al.*[43] performed a $\approx$ 1 ms long all-atom equilibrium MD simulation at 300 K, using the AMBER ff99SB force field[60] and the TIP4P-Ew[64] water model. From this extensive simulation we adopted a $\approx$62.5 μs long representative segment of frames, with a total of 250 000 snapshots sampled at every 250 ps.

C. Rotational fit

To determine the overall rotation matrix in Eq. (2), we employed Kabsch's method[47] as implemented in the GROMACS 4.6.1 program suite,[65] which minimizes

$$\Delta = \sum_{i=1}^{N} m_i (\mathbf{r}'_i - \bar{\mathbf{r}}_i)^2, \tag{3}$$

i.e., the least-square distance between the mass-weighted instantaneous and reference structures $\mathbf{r}' = \{\mathbf{R}^T \mathbf{r}_i\}$ and $\bar{\mathbf{r}}$, respectively. For HP35, we used several reference structures, including the native structure obtained from experiment[66] and a native-like metastable state identified in a dynamical clustering analysis[67] of the considered MD trajectory. For BPTI, only the native structure obtained from x-ray scattering[68] was employed.

Apart from this standard procedure, we applied two recently proposed advanced fitting schemes. To avoid ambiguous rotations, where different rotation matrices are applied to similar structures, Gapsys and de Groot developed various algorithms which aim to minimize the variance of the structural ensemble as well as distance of each structure to preceding MD frames or its nearest neighbors.[55] While the approach could not improve the situation for HP35, it made a difference for BPTI, for which the "min(Var+NN)" scheme with NN = 25 nearest neighbors was employed. Moreover, we used the multi-reference fitting method proposed in Ref. 69, which fits the structure at time t_n to the structure of its preceding MD frame at t_{n-1}. The idea is that consecutive MD structures should be very similar to each other and therefore more suitable for a proper fit than arbitrarily chosen structures from the same trajectory. Employing nonequilibrium MD simulations of the vibrational energy redistribution in a small peptide,[69] it has been shown that the approach correctly separates internal and rotational energies, as can also be done by the method of Jellinek and Li.[57] However, as every fit introduces some error, the procedure results in overall structural deviations that accumulate in the course of the simulation, see Fig. S1 in the supplementary material.[76]

D. Principal component analysis

The correlated internal motion of a molecule with N atoms can be described by a covariance matrix,

$$\sigma_{ij} = \langle (r_i - \langle r_i \rangle)(r_j - \langle r_j \rangle) \rangle, \tag{4}$$

where $r_1, \ldots, r_{3N}$ are mass-weighted Cartesian coordinates and $\langle \ldots \rangle$ represents the average over all sampled conformations. Diagonalization of this covariance matrix results in 3N eigenvectors ($\mathbf{v^i}$) and eigenvalues ($\lambda_\mathbf{i}$) which describe the modes of the collective motion and their respective amplitudes. The principal components,

$$\mathbf{V_i} = \mathbf{v^i} \cdot \mathbf{r} \tag{5}$$

are the projections of the data $\mathbf{r} = \left(r_1, \ldots, r_{3N}\right)^T$ onto the eigenvectors, and can be used to construct low-dimensional free energy profiles of the system. Using the first two principal components, for example, we obtain the two-dimensional free energy landscape

$$\Delta G(V_1, V_2) = -k_\mathrm{B} T \ln P(V_1, V_2), \tag{6}$$

where P is the joint probability density function of the data along the two principal components.

Instead of Cartesian coordinates, one may also use internal coordinates such as the backbone dihedral angles (ϕ, ψ). Being circular variables, however, the angles first need to be transformed to a linear *metric* coordinate space (i.e., a vector space with the usual Euclidean distance). This can be achieved by the transformation[23]

$$\begin{aligned} q_{2n-1} &= \cos \varphi_n, \\ q_{2n} &= \sin \varphi_n, \end{aligned} \tag{7}$$

where $n = 1, \ldots, M$ and M is the total number of peptide backbone (and possibly also side-chain) dihedral angles used in the analysis. This doubling of variables occurring in the dPCA can be explained by considering a complex version of the theory using $z_n = e^{\mathrm{i}\varphi_n}$, which also showed that dPCA amounts to a one-to-one representation of the original angle distribution.[37] In particular, this means that the dPCA cannot lead to additional spurious minima of the free energy

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surface,[70] and that two distinct minima on the dPCA free energy surface must correspond to conformational states of different structure.[71] Details of the PCAs on HP35 are given in Ref. 67. For BPTI, the supplementary material shows the convergence of the cumulative fluctuations and the free energy curves along the first few principal components for a dPCA and a cPCA with standard and min(Var+NN) fitting.

E. Most probable path clustering

In order to identify metastable conformational macrostates in the principal subspace, we employ kinetic clustering using the *most probable path* algorithm introduced recently.[72] First, the *k*-means geometric clustering algorithm[73] is used in an unsupervised manner to reduce the large number of MD snapshots ($\sim 10^6$–10^7) to a manageable number of microstates ($\sim 10^3$–10^4). The resulting microstates can then be merged into metastable states using the most probable path algorithm, assuming a time-scale separation between the fast intrastate motion within the metastable states and slow interstate transitions. The algorithm computes the transition matrix $\{T_{ij}\}$ for the set of microstates, where T_{ij} represents the probability of a state i to change to state j within a predefined lag time (τ) and the self-transition probability T_{ii} represents the metastability of state i. Starting from a given microstate, the most probable path algorithm calculates the transition probabilities of this state and chooses the most probable transition for all states with $T_{ii} \leq Q_{\min}$, where $Q_{\min}$ is a chosen minimum value of the metastability. This is done iteratively, until we reach a state i with $\max(T_{ij}) \leq T_{ii}$, i.e., it is most likely to stay in this state. Since, *intra*-basin transitions are much more likely than *inter*-basin transitions (i.e., barrier crossing), this algorithm collects all the microstates of a basin, thereby defining the basin in terms of the included microstates. Moreover, this approach by construction places the boundaries in the middle of the barriers separating the metastable states.

The dynamical clustering of the trajectory of HP35 was described in detail in Ref. 67. In the case of BPTI, we generated 5000 microstates via *k*-means geometrical clustering and performed the most probable path clustering. Choosing $Q_{\min} = 0.98$, we found six distinct metastable states of BPTI (see Fig. S7 in the supplementary material[76]), whose structures are shown in Fig. 4.

III. RESULTS AND DISCUSSION

A. Large amplitude motion: The folding of HP35

Applying the most probable path algorithm to a long MD trajectory of villin headpiece (HP35) at 380 K,[42] a recent study has shown that the folding free energy landscape of HP35 consists of three main basins, accounting for the unfolded (U), the intermediate (I) and the native (N) states, respectively.[67] The unfolded basin U was found to be of entropic nature, while the basins I and N each contain two highly populated substates, indicating the enthalpic nature of these basins. To get a first impression, Fig. 1 shows the free energy landscape ΔG along the first two principal components

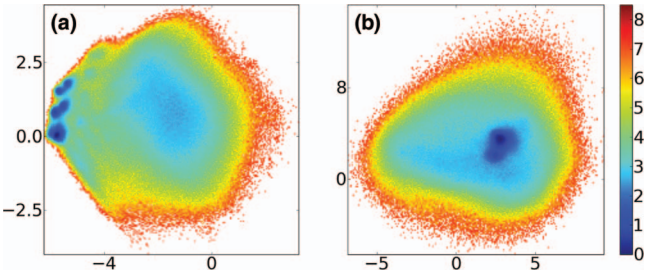


FIG. 1. Free energy landscape $\Delta G(V_1, V_2)$ (in units of $k_B T$) of the folding of HP35 obtained from a PCA using (a) the backbone dihedral angles of the protein (dPCA) and (b) the Cartesian coordinates of all backbone atoms (cPCA), respectively.

V_1 and V_2 of a PCA using (a) the backbone dihedral angles of the protein (dPCA) and (b) the Cartesian coordinates of all backbone atoms (cPCA), respectively. Similar as found in the dynamical clustering analysis of Ref. 67, the dPCA landscape in (a) reveals a broad entropic unfolded basin U, which is separated by a barrier from several well-defined native-like states, which are the substates of the intermediate and native basins I and N. In contrast, the cPCA landscape in (b) exhibits a simple funnel-like shape and contains only a single minimum but no barrier. Although it is clear that low-dimensional projections often yield only limited information on the multidimensional energy landscape,[38,74] this qualitative behavior was found quite generally, e.g., for any combinations of principal components. That is, the cPCA appears to fail completely to characterize even the qualitative structure of the folding energy landscape of HP35.

To explain the origin of this breakdown, we first explored different versions of cPCA which employ alternative fitting procedures (see Sec. II). As an example, Fig. S1 in the supplementary material[76] shows the results of a cPCA using (i) a different reference structure, (ii) the multireference method,[69] and (iii) the min(Var+NN) method,[55] respectively. However, all variants gave similar single-minimum-type energy landscapes for HP35.

On second thought, we analyzed the problem from a *local* perspective. That is, we start with the structural analysis of a small part of the protein, for which both dPCA and cPCA should give the same results.[54] In a second step, we increase the size of the considered subsystem in order to learn at what point and why the problem arises. In the case of HP35, we selected as a small subsystem a single residue, ALA8, which undergoes considerable conformational fluctuations. This can be seen from the corresponding (ϕ, ψ) Ramachandran plot in Fig. 2(a) which reveals the coexistence of four conformational states, α_R, β, P_{II}, and α_L. Being defined by only two dihedral angles, a dPCA using two principal components trivially gives the correct conformational distribution of ALA8 (data not shown).

In the case of a cPCA that uses the Cartesian coordinates of the five backbone atoms of ALA8, on the other hand, the outcome depends crucially on the rotational fit preceding the cPCA. If the fit is performed with respect to only these five atoms (henceforth referred to as "*local fit*"), the resulting cPCA energy landscape in Fig. 2(b) also reveals four well-defined minima, which can directly be assigned to the

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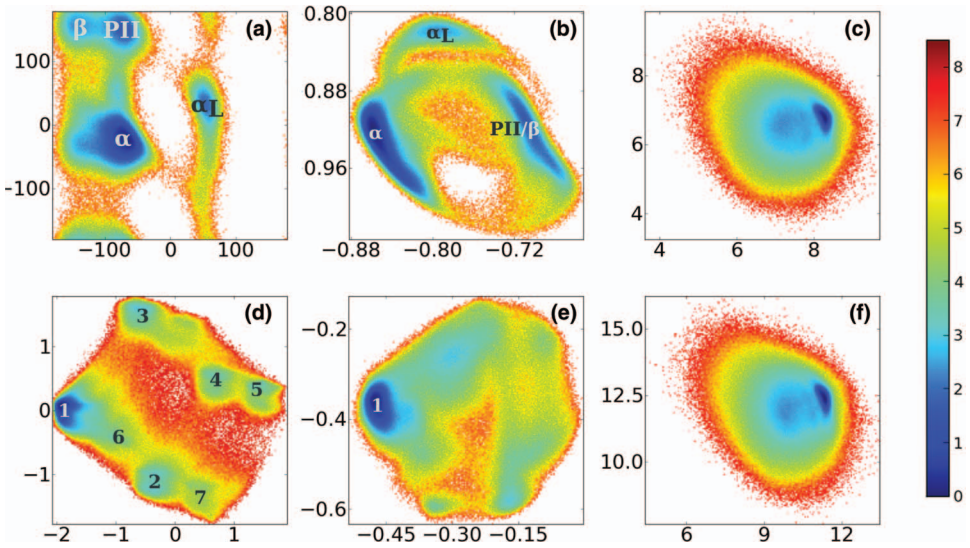


FIG. 2. (Top) Free energy landscape (in units of $k_B T$) of a single residue (ALA8) of HP35 as (a) found along the (ϕ, ψ) dihedral angles as well as obtained from a cPCA (projected on V_1 and V_2) after (b) local and (c) global fitting. (Bottom) Similar as above, but for three adjacent residues (LYS7, ALA8, and VAL9) of HP35. The dPCA energy landscape in (d) reveals seven metastable conformational states, that can be described by their state number, secondary structure, and population: **1** ($\alpha\alpha\alpha$) 56%, **2**($\alpha\beta\alpha$) 8%, **3** ($\alpha\alpha\beta$) 7%, **4** ($\alpha\beta\beta$) 5%, **5** ($\beta\beta\beta$) 4%, **6** ($\alpha\alpha_L\alpha$) 3%, and **7** ($\beta\beta\alpha$) 3%. The locally fitted cPCA in (e) recovers only the highest populated state **1**, while the structures associated with the minimum of the globally fitted cPCA in (d) are completely mixed.

metastable states of the system. Hence a locally fitted cPCA of ALA8 reproduces the correct conformational distribution. The picture changes completely, however, if we perform the rotational fit with respect to *all* atoms of HP35 ("*global fit*"). Figure 2 shows that a globally fitted cPCA of ALA8 exhibits a diffuse single-minimum energy landscape, quite similar to the cPCA performed for the complete system in Fig. 1(b). A further significant difference arising from the two fitting schemes is the variance of the cPCA data, which is about a factor 30 larger in the case of the global fit. This indicates that the conformational distribution given by the globally fitted cPCA mostly reflects the dominant *overall* motion rather than the much smaller *internal* motion of the folding protein. We stress that the local fit is certainly the correct way to subtract the overall rotation of a single residue. Nonetheless, we have to employ the approximation of a global fit for a cPCA of the complete system, of which ALA8 is a part.

To study how these findings change when we include a larger part of the protein in the analysis, we next considered three consecutive residues, LYS7, ALA8, and VAL9. A dPCA on the backbone dihedral angles of these residues reveals seven conformational states, which directly correspond to the seven most populated states identified in a cluster analysis (Fig. 2(d)). When we perform a cPCA with a local fit including all backbone atoms of the three residues, on the other hand, we only recover the highest populated state **1**. The other diffuse features of the energy landscape cannot be assigned to the remaining metastable states but correspond to random mixtures of conformations. When we perform a globally fitted cPCA, we obtain again the spurious single-minimum energy landscape with very large variance, see Fig. 2(f). These findings clearly indicate that the separation of overall and internal motion deteriorates with increasing number of flexible residues included in the fit. As a consequence, a subsequent cPCA is corrupted by the dominant overall motion, which manifests itself in a spurious energy landscape.

To introduce a measure for the deviation of a fitted structure from the reference structure, we consider the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ along the $N - C_\alpha$ bond of a selected residue in the fitted and the reference structure, respectively, and define the deviation between reference and fit by the angle

$$\theta = \arccos\left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{|\mathbf{v}_1|\,|\mathbf{v}_2|}\right). \tag{8}$$

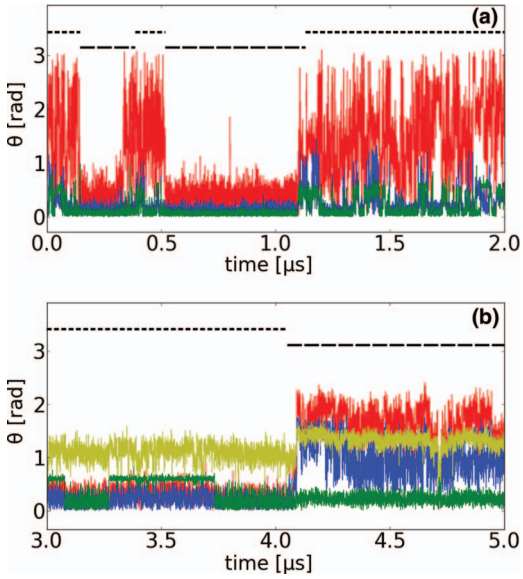


FIG. 3. Time evolution of the angular deviation θ defined in Eq. (8), pertaining to (a) residue ALA8 of HP35 and (b) residue ARG17 of BPTI. Shown are results obtained from a local fit (green), a fit against the backbone atoms of the neighboring residues (blue), and a global fit using the standard scheme (red) as well as the min(Var+NN) method (yellow, shown for (b) only). In (a), the dashed lines indicates if HP35 populates the unfolded basin (upper lines) or the native and intermediate basins (lower lines). In (b), the dashed line indicates if BPTI occupies a conformation of the reference state (upper lines) or another metastable state (lower lines).

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Figure 3(a) shows the time evolution of the resulting angular deviation of residue ALA8 in the case of a local fit (green), a fit against the backbone atoms of the three consecutive residues 7, 8, and 9 (blue), and a global fit against all backbone atoms (red). Using a local fit, we find only relatively small deviations, with θ fluctuating either around 0.06 or around 0.47. These values of θ correspond to the two dominant conformations α_R and $\beta/\mathrm{P_{II}}$ of ALA8 (cf. Fig. 2(a)). They reflect the fact that we necessarily obtain a non-perfect fit with $\theta \neq 0$, since we use a single reference structure to fit all conformations of ALA8. Fitting against three consecutive residues, we need to account for all eight possible conformational states of these residues (cf. Fig. 2(d)) by a single reference structure. While this requirement necessarily leads to larger angular deviations, the time evolution of θ is found to still monitor the two dominant conformations α_R and $\beta/\mathrm{P_{II}}$ of ALA8. Finally, when we include all 35 flexible residues of HP35 in the fit, a single reference structure is certainly not appropriate any more to match all the structures of the folding protein, regardless which reference structure or fitting method (standard or min(Var+NN)) is used. This leads to large angular deviations which are not correlated any more with the conformational states of ALA8.

The analysis clearly shows that the above found breakdown of the cPCA for the folding of HP35 is actually a failure of the preceding rotational fit. Fitting a molecule that undergoes large-amplitude motion against a single reference structure cannot eliminate the overall rotation, because the rotation depends via the moment of inertia on the molecule's structure, which is changing drastically. This necessarily results in a mixing of overall and internal motion such that the conformational distribution given by the globally fitted cPCA mostly reflects the dominant *overall* motion rather than the much smaller *internal* motion of the folding protein. We note that this problem occurs only if the trajectory populates the unfolded basin of the energy landscape (indicated by the dashed line in Fig. 3(a)). In the native and intermediate basins (e.g., for $0.5\ \mu\mathrm{s} \leq t \leq 1\ \mu\mathrm{s}$), on the other hand, HP35 remains in a native-like overall structure and ALA8 stays in a single conformational state. This results in comparatively small fluctuations of θ, even in the case of a global fit.

B. Small-amplitude motion: Functional dynamics of BPTI

In contrast to the large-amplitude motion of protein folding, the function of proteins is often associated with small-amplitude motion. In this case, a rotational fit using a single reference structure is expected to be sufficient to eliminate the overall rotation of the system, and therefore a cPCA should work and give similar results as a PCA using internal coordinates. To explore this thought, we consider the functional dynamics of BPTI, as inferred from a long MD trajectory of Shaw *et al.*[43] (see Sec. II). Due to three disulfide bridges and a well-defined hydrophobic core, the 58-residue protein BPTI is known to be quite stable.[68] Nonetheless, an analysis of the MD simulation in Ref. 43 revealed five metastable conformational states, two of which were already known from experiment.[75] Here we performed a most probable path dynamical clustering analysis (see Sec. II), which resulted in (at least) six conformational states. As shown in Fig. 4, the states differ mostly in the conformations of the first and third flexible loop of BPTI shown in red, while the two β-sheets (yellow), and the α-helix (blue) remain stable.

To investigate to what extend a PCA can resolve the conformational distribution of BPTI, Fig. 4 compares the free energy landscapes obtained by (a) a dPCA as well as by a cPCA on the backbone atoms, using (b) standard and (c) min(Var+NN) fitting, respectively. The dPCA free energy along the first seven principal components is quite structured and exhibits various well-defined minima (see the supplementary material[76]). Projected on the first two principal components, the energy landscape in Fig. 4(a) indicates numerous conformational states. The two versions of the cPCA, on the other hand, show only three principal components with a structured distribution. Projecting on the two most structured components V_1 and V_3, the cPCA energy landscapes in Figs. 4(b) and 4(c) show four and six conformational states, respectively.

To assign the features of the energy landscapes to the states obtained from the most probable path dynamical clustering analysis, panels (d)–(f) of Fig. 4 show the projection of these states on the two-dimensional subspaces of the various PCAs. Panel (d) reveals that the dPCA clearly discriminates the six main metastable conformational states. Moreover, the dPCA identifies several substates that are found by the most probable path dynamical clustering analysis when a smaller metastability is required (see Fig. S8 in the supplementary material[76]). The cPCAs as well provide a correct representation of the energy landscape, in the sense that all states are comprised in some minimum of the energy surface. The main difference compared to the dPCA is the minor state resolution, in particular in the case of standard fitting. The min(Var+NN) fit, on the other hand, represents a clear improvement of the state resolution.

To test if the lower resolution of the cPCA is caused by a fitting problem (as was the case in the folding of HP35), we again adopt a local perspective and consider the conformational distribution of a single flexible residue (ARG17), see Fig. 5. The corresponding Ramachandran plot in panel (a) reveals the coexistence of four conformational states, α_R, β, $\mathrm{P_{II}}$, and α_L, which are nicely reproduced by a locally fitted cPCA in panel (b). In the case of the small-amplitude motion of BPTI, moreover, also a globally fitted cPCA is able to discriminate the main conformational states. In this case, standard and min(Var+NN) fitting give almost identical results. However, the overall resolution is reduced (α_L and β are mixed, $\mathrm{P_{II}}$ is missing) and the variance of the cPCA data in (c) is again significantly larger than in the case of (b).

As for HP35, we extended the test to a few adjacent residues (CYS14, LYS15, ALA16, ARG17). The dPCA in Fig. 5(d) discriminates eight states that differ by their secondary structure (see caption). The locally fitted cPCA landscape is similar, but combines states **6** and **7** in a single state and misses state **8**. The globally fitted cPCA landscape still recovers four of the eight states. Again the results from standard and min(Var+NN) fitting are very similar. The decrease of state resolution of the globally fitted energy

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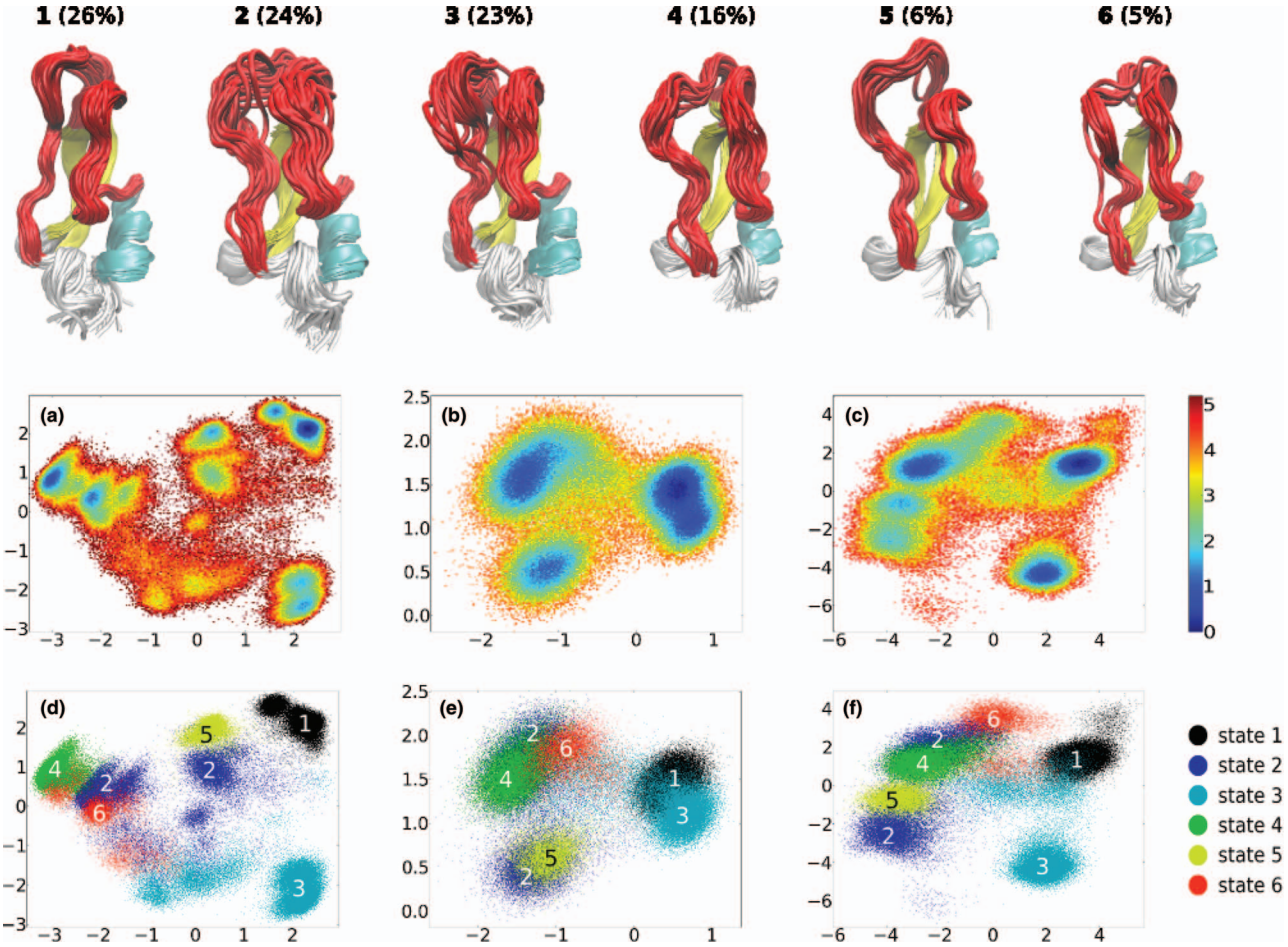


FIG. 4. (Top) Representative molecular structures of the six main metastable states of BPTI with their thermal population given in parentheses, as obtained by the most probable path clustering. (Middle) Free energy landscape of BPTI, obtained by (a) a dPCA and ((b) and (c)) a cPCA with (b) standard and (c) min(Var+NN) fitting, respectively. (Bottom) Conformational distribution of the metastable states shown above, projected on the two-dimensional subspace of the above-lying PCA.

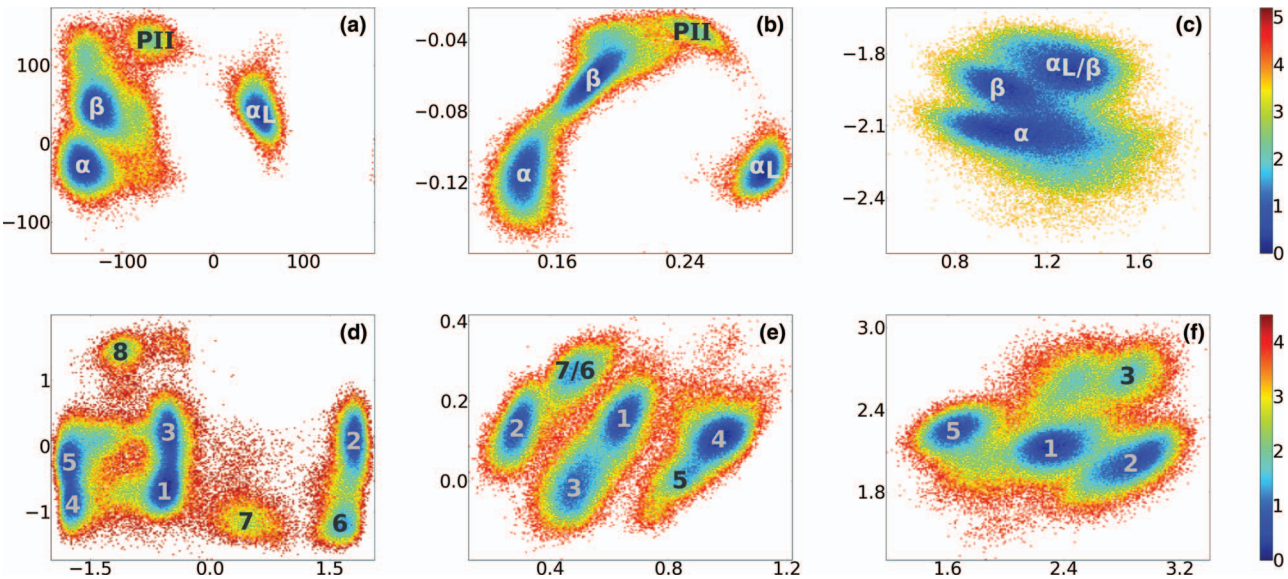


FIG. 5. (Top) Free energy landscape associated with residue ARG17 of BPTI, as given by the corresponding Ramachandran (ϕ, ψ) plot (a), and by a cPCA after local (b) and global (c) fitting, respectively. (Bottom) Similar as above, but for the four adjacent residues CYS14, LYS15, ALA16, and ARG17. The dPCA energy landscape in (d) indicates eight metastable conformational states: **1** ($P_2\alpha P_2\beta$), **2** ($\beta\alpha\alpha\alpha_L$), **3** ($\beta\alpha\beta\beta$), **4** ($P_2\beta\beta\alpha$), **5** ($\beta\beta\beta\alpha$), **6** ($P_2\alpha\alpha\alpha_L$), **7** ($P_2\alpha P_2 P_2$), **8** and ($\alpha\alpha_L P_2\alpha$), which are partly also found the cPCAs in (e) and (f).

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landscape compared to the locally fitted landscape clearly shows that the separation of global and internal motion is not perfect, even in the case of the small-amplitude functional motion of BPTI.

To further illustrate this finding, we again consider the angular deviation θ of a specific residue for various fitting methods. Figure 3(b) shows the time evolution of θ of residue ARG17 in the case of a local fit, a fit against the backbone atoms of the four consecutive residues 14-17, and a global fit using the standard method as well as the min(Var+NN) scheme. As expected, the local fit shows relatively small deviations. Interestingly, the four-residue fit as well as the standard global fit yield large fluctuations after $t \approx 4\mu$s. At this time, BPTI changes from a conformation similar to the reference structure the fit was performed on to another metastable state of BPTI. When we employ a min(Var+NN) fit, this effect becomes smaller, which indicates that the large fluctuations of $\theta(t)$ obtained by the standard fit are caused by ambiguous rotations.

IV. CONCLUDING REMARKS

Considering the structural dynamics of flexible non-isolated biomolecules, we have studied the qualitative features of the resulting free energy landscape obtained from a principal component analysis (PCA). We considered two representative cases of protein dynamics, the large-amplitude folding of HP35 and the small-amplitude functional motion of BPTI, and employed in the analysis Cartesian coordinates (cPCA) as well as internal coordinates (dPCA). The folding of a small and flexible protein represents a worst-case scenario of the separation problem, as rotational and internal motion is intrinsically coupled through the time-dependent inertia tensor. Independently of the rotational fitting scheme employed, we found that a cPCA completely fails to reproduce even the qualitative features of the folding free energy landscape of HP35. Hence one can clearly not advice to use a cPCA to describe large-amplitude conformational dynamics of flexible peptides and proteins and other intrinsically disordered systems.

In the case of the functional motion of BPTI, on the other hand, both cPCA and dPCA were able to distinguish the main metastable states of the system (Fig. 4). However, even in this case of overall small-amplitude motion, the cPCA achieved only an approximate representation of the energy landscape, showing minor resolution and significantly higher variance. The analysis provided in Figs. 3 and 5 showed that this lack of resolution is to a large extent caused by fitting problems of the cPCA. In the case of small-amplitude dynamics, however, advanced fitting scheme such as the min(Var+NN) method[55] may improve the separation problem.

While we restricted ourselves in this work to two examples, the analysis suggests that our findings associated with the separation problem generalize to most flexible biopolymers. Furthermore, it should be stressed that this problem occurs not only in the construction of free energy landscape, but in most applications of a cPCA. For example, the separation problem renders the rigorous interpretation of long-range correlations between distant residues questionable.[45,48,49] Moreover, Cartesian trajectory snapshots are generally not suited to construct microstates for a clustering analysis, unless the overall rotation is removed separately for each pair of microstates.[5,67] This note of caution generalizes to most kinds of data preprocessing a PCA is often used for.

Cartesian coordinates are in general more convenient to handle, e.g., for the visualization of the molecular structure. Due to the separation problem, on the other hand, internal coordinates often give a better representation of the protein's energy landscape. In particular, a general advantage of the dPCA is that it focuses on the relevant internal degrees of freedom and thus avoids unnecessary noise. That is, it disregards the remaining internal degrees of freedom (bond lengths and bond angles), which undergo fast fluctuations but are not of interest for the structural dynamics of the system. While backbone dihedral angles have shown to be useful for relatively small biomolecules, it has been noted that the motion of adjacent backbone dihedral angles is necessarily correlated in a protein with relatively rigid secondary structures (the "loop closure problem").[41,51] This nonlinear correlation may render dihedral angles an inefficient representation of the dynamics, in particular when large-amplitude functional motion of proteins is considered. In these cases, other choices of internal coordinates such as contacts or residue-residue distances[32–34] might be advantageous.

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[76]See the supplementary material at http://dx.doi.org/10.1063/1.4885338 for details on various fitting procedures for HP35 (Fig. S1), several PCAs for BPTI (Figs. S2-S6), and the clustering of BPTI (Figs. S7 and S8).

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